Assignment 7 Solutions

1.1 $\lambda x.xx$

1.2 $(\lambda x.x)(\lambda x.x)\,x$

1.3 $(\lambda x.\lambda y.yx)(\lambda x.x)(\lambda z.zz)$

1.4 $\lambda x.\,xxy\,(yzw)\,x\,(\lambda x.xyz)$

2.1 $\lambda x.((xx)x)$

2.2 $\lambda x.(\lambda y.((yz)(\lambda x.x)))$

2.3 $((xy)(zw))(\lambda x.(\lambda x.(x(\lambda x.x))))$

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$$\frac{}{(\lambda x.xx)(\lambda y.y) \longrightarrow (\lambda y.y)(\lambda y.y)}\ (\beta)^{①}$$

① $[\lambda y.y / x](xx)$
$= ([\lambda y.y / x]x)([\lambda y.y / x]x)$
$= (\lambda y.y)(\lambda y.y)$

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$$\frac{\dfrac{}{(\lambda y.y)(\lambda y.y) \longrightarrow \lambda y.y}\ \beta^{②} \qquad \dfrac{}{\lambda y.y \longrightarrow^{*} \lambda y.y}\ \text{refl}}{(\lambda y.y)(\lambda y.y) \longrightarrow^{*} (\lambda y.y)}\ \text{step}$$

$$\frac{\dfrac{}{(\lambda x.xx)(\lambda y.y) \longrightarrow (\lambda y.y)(\lambda y.y)}\ (\beta)}{(\lambda x.xx)(\lambda y.y) \longrightarrow^{*} (\lambda y.y)}\ \text{step}$$

② $[\lambda y.y / y]\,y$
$= \lambda y.y$

6

$$\frac{\dfrac{}{(\lambda y.y)(\lambda z.z) \longrightarrow \lambda z.z}\ \beta^{③}}{(\lambda x.x)((\lambda y.y)(\lambda z.z)) \longrightarrow (\lambda x.x)(\lambda z.z)}\ \text{appRight}$$

③ $[\lambda z.z / y]\,y$
$= \lambda z.z$

$$\frac{}{(\lambda x.x)((\lambda y.y)(\lambda z.z)) \longrightarrow (\lambda y.y)(\lambda z.z)}\ \beta^{④}$$

④ $[(\lambda y.y)(\lambda z.z) / x]\,x =$
$(\lambda y.y)(\lambda z.z)$

9

$$\frac{\dfrac{\dfrac{}{(\lambda y.x)(\lambda x.xx) \longrightarrow x}\ \beta^{⑤}}{(\lambda x.(\lambda y.x)(\lambda x.xx)) \longrightarrow \lambda x.x}\ \text{fun}}{(\lambda x.(\lambda y.x)(\lambda x.xx))(\lambda x.yx) \longrightarrow (\lambda x.x)(\lambda x.yx)}\ \text{appLeft}$$

⑤ $[\lambda x.xx / y]\,x$
$= x$

10.1

$$\begin{aligned} FV(\lambda x.xy) &= FV(xy) \setminus \{x\} \\ &= (FV(x) \cup FV(y)) \setminus \{x\} \\ &= (\{x\} \cup \{y\}) \setminus \{x\} \\ &= \{x, y\} \setminus \{x\} \\ &= \{y\} \end{aligned}$$

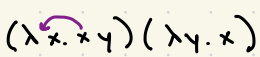
$$\begin{aligned} BV(\lambda x.xy) &= BV(xy) \cup \{x\} \\ &= (BV(x) \cup BV(y)) \cup \{x\} \\ &= \emptyset \cup \{x\} \\ &= \{x\} \end{aligned}$$



10.2

$$\begin{aligned} FV((\lambda x.xy)(\lambda y.x)) &= FV(\lambda x.xy) \cup FV(\lambda y.x) \\ &= \{y\} \cup (FV(x) \setminus \{y\}) \\ &= \{y\} \cup (\{x\} \setminus \{y\}) \\ &= \{y\} \cup \{x\} \\ &= \{x, y\} \end{aligned}$$

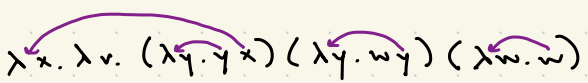
$$\begin{aligned} BV((\lambda x.xy)(\lambda y.x)) &= BV(\lambda x.xy) \cup BV(\lambda y.x) \\ &= \{x\} \cup (BV(x) \cup \{y\}) \\ &= \{x\} \cup \{y\} \\ &= \{x, y\} \end{aligned}$$



10.3

$$
\begin{aligned}
&FV(\lambda x.\lambda v.(\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \\
&= FV(\lambda v.(\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \setminus \{x\} \\
&= (FV((\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \setminus \{v\}) \setminus \{x\} \\
&= (FV((\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \setminus \{x,v\} \\
&= (FV(\lambda y.yx) \cup FV(\lambda y.wy) \cup FV(\lambda w.w)) \setminus \{x,v\} \\
&= (FV(\lambda y.yx) \cup FV(\lambda y.wy) \cup (FV(w) \setminus \{w\})) \setminus \{x,v\} \\
&= ((FV(yx) \setminus \{y\}) \cup (FV(wy) \setminus \{y\}) \cup (\{w\} \setminus \{w\})) \setminus \{x,v\} \\
&= ((\{y,x\} \setminus \{y\}) \cup (\{w,y\} \setminus \{y\}) \cup \emptyset) \setminus \{x,v\} \\
&= (\{x\} \cup \{w\}) \setminus \{x,v\} \\
&= \{x,w\} \setminus \{x,v\} \\
&= \{w\}
\end{aligned}
$$

$$
\begin{aligned}
&BV(\lambda x.\lambda v.(\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \\
&= BV((\lambda y.yx)(\lambda y.wy)(\lambda w.w)) \cup \{x,v\} \\
&= ((BV(yx) \cup \{y\}) \cup (BV(wy) \cup \{y\}) \cup (BV(w) \cup \{w\})) \cup \{x,v\} \\
&= \{y,w\} \cup \{x,v\} \\
&= \{x,y,v,w\}
\end{aligned}
$$



13.1

$$
\begin{aligned}
[\lambda z.z/x](xyz) &= ([\lambda z.z/x](xy))([\lambda z.z/x]z) \\
&= (([\lambda z.z/x]x)([\lambda z.z/x]y))z \\
&= (\lambda z.z)yz
\end{aligned}
$$

13.2

$$
\begin{aligned}
[\lambda z.zz/x](\lambda y.x(\lambda x.xy)) &= \lambda y.[\lambda z.zz/x](x(\lambda x.xy)) \\
&= \lambda y.([\lambda z.zz/x]x)([\lambda z.zz/x](\lambda x.xy)) \\
&= \lambda y.(\lambda z.zz)(\lambda x.xy)
\end{aligned}
$$

13.3

$$
\begin{aligned}
[\lambda y.y/x]((\lambda x.xy)(\lambda y.x)) &= ([\lambda y.y/x](\lambda x.xy))([\lambda y.y/x](\lambda y.x)) \\
&= (\lambda x.xy)(\lambda y.[\lambda y.y/x]x) \\
&= (\lambda x.xy)(\lambda y.\lambda y.y)
\end{aligned}
$$

15.1

$$
\begin{aligned}
[yz/x](\lambda w.xw) &= \lambda w.[yz/x](xw) \\
&= \lambda w.([yz/x]x)([yz/x]w) \\
&= \lambda w.yzw
\end{aligned}
$$

15.2

$$
\begin{aligned}
[y/x](\lambda y.xy) &= \lambda w.[y/x][w/y](xy) \\
&= \lambda w.[y/x](([w/y]x)([w/y]y)) \\
&= \lambda w.[y/x](xw) \\
&= \lambda w.([y/x]x)([y/x]w) \\
&= \lambda w.yw
\end{aligned}
$$

15.3

$$
\begin{aligned}
&[xy/z](\lambda x.z(\lambda z.z)(\lambda w.zx)) \\
&= \lambda q.[xy/z][q/x](z(\lambda z.z)(\lambda w.zx)) \\
&= \lambda q.[xy/z](([q/x]z)([q/x](\lambda z.z))([q/x](\lambda w.zx)) \\
&= \lambda q.[xy/z](z(\lambda z.[q/x]z)(\lambda w.[q/x](zx))) \\
&= \lambda q.[xy/z](z(\lambda z.z)(\lambda w.([q/x]z)([q/x]x))) \\
&= \lambda q.[xy/z](z(\lambda z.z)(\lambda w.zq)) \\
&= \lambda q.([xy/z]z)([xy/z](\lambda z.z))([xy/z](\lambda w.zq)) \\
&= \lambda q.(xy)(\lambda z.z)(\lambda w.[xy/z](zq)) \\
&= \lambda q.(xy)(\lambda z.z)(\lambda w.([xy/z]z)([xy/z]q)) \\
&= \lambda q.(xy)(\lambda z.z)(\lambda w.xyq)
\end{aligned}
$$